

Subject code and title: DLE602 Deep Learning

Subject length: 6 weeks and 12 weeks

Subject Description:

This subject introduces theories of Deep learning and how to build neural networks. Deep learning is a subset of machine learning, and machine learning is a subset of Artificial Intelligence (AI). Students are presented with different methods and applications of different AI sub-areas such as natural language processing, speech recognition and computer vision. The subject progresses from the introduction of neural networks such as multi-layer perceptron to complex concepts such as recurrent neural networks, convolutional networks and long short- term networks.

Subject Learning Outcomes (SLO):

SLO a)	Build, train and apply deep learning models to real- world tasks.
SLO b)	Compare and select ways to pre-process signals, images, and texts for natural language, speech recognition, and computer vision applications.
SLO c)	Develop critical analysis skills in deep learning research.
SLO d)	Demonstrate collaborative skills to apply deep learning to solve real world problems.
SLO e)	Demonstrate ability to effectively communicate scientific and technical information.

Delivery Schedule:

	Module	Learning Activities	Assessment Progression & Due Date	SLOs addressed
1	Introduction to Deep Learning & Neural network	- Discussion forum: Introduce Yourself - Discussion forum: (Assessment 2 & 3 Preparation)	Familiarise yourself with Assessment 1, understand key concepts of deep learning and review algorithms like N-Gram in preparation for Assessment 1.	a
2	Deep Learning: Feedforward Neural Network and Backpropagation	- Discussion forum: Understanding Feedforward Network - Discussion forum: Interactive Knowledge Sharing - Backpropagation - Discussion forum: The Disadvantage of Neural Networks	Understand the basic concept of neural networks and their associated algorithms, identify and practise ways to convert algorithms into programming implementation using Python; and continuing to review the research paper suggested for Assessment 1.	a, b
3	Deep Learning Application: Natural Language Processing, Speech Recognition and Computer Vision	- Discussion forum: Natural Language Processing - Discussion forum: Speech Recognition - Discussion forum: Computer Vision	Assessment 1 progression: Gain a better understanding about the non-functional requirements of your programming solution based on the applications explored in this Module Assessments 2 and 3 progression: Review the research papers in this Module and acquire a better sense of what can and cannot be proposed for your project.	a, b, d
4	Regularization	- Discussion forum: Programming - Discussion forum: Critical analysis - Discussion forum: I Wish	Assessment 1 due: Programming Problems (30%)	a, c, d
5	Convolutional Neural Networks	- Discussion forum: Critical Analysis I – Initial Recommendation - Discussion forum: Critical Analysis II– Final Recommendation - Discussion forum: Train the Network - Wiki Activity: Collaboration	Identify possible use of convolutional neural networks, analyse specific convolutional neural network algorithms and look at some Python implementation of convolutional neural networks.	a, b, c, d

6	Linear Factor Models	• Discussion forum: Comparison • Discussion forum: Idea Generation • Discussion forum: Open Forum	Identify possible use of linear factor models for your projects and dig deep into your possible input dataset and understand the data for the rest of the project.	b, c, d, e
7	Autoencoders	• Discussion forum: Comparison • Discussion forum: Fake to Real • Discussion forum: Deal or no-deal	Identify possible use of autoencoders for your projects and dig deep into your possible input dataset, particularly for assessment 3, and understand the data for the rest of the project.	c, d, e
8	Recurrent Neural Networks and Long Short-Term Memory	• Discussion forum: Comparison • Discussion forum: What Do You Think? • Discussion: Applications	Assessment 2 due: Deep Learning Project Proposal Presentation (30%)	c, d, e
9	Representation Learning	• Discussion: Warm up • Discussion forum: Think Before You Write • Discussion forum: Applications	Take a close look at the dataset, analyse what is the best way to represent your input data and consider the impact that using the concept of representation learning may have on your project.	c, e
10	Structured Probabilistic Models for Deep Learning	• Discussion forum: Directed versus Undirected • Discussion forum: Advantages of Structured Modelling • Discussion forum: Graphical Models for COVID-19	Consider about the usage of structured and unstructured modelling for Assessment 3, if applicable.	c, e
11	Visual Analytics in Deep Learning	• Discussion forum: Opinion Matters • Discussion forum: Interactive Learning Activity • Discussion	Exploring visual analytics, if your project could benefit from it and read the essential resources as they contain some critical information for you to better explain, interpret, debug and improve the neural networks in your project.	c, d, e
12	Practical Methodologies and the Future of Deep Learning	• Discussion forum: Last Opinion • Discussion forum: The Future is Here • Discussion	Assessment 3 due: Deep Learning Final Project (40%)	c, e

Time Management:

Below outlines the minimum time management expectations for students.

6 Week Delivery:

- 10 Hours per module: Facilitated study: 3 hours / module. Personal Study: 7 hours / module. (Typically 2 modules per week).
- 6 hours facilitated study consists of attending class, responding to facilitator feedback.
- Students are to allocate 14 hours of personal learning. This includes essential time spent on pre-reading and viewing materials, assessment progression and learning activities.

12 Week Delivery:

- 10 Hours per module (one week): Facilitated study: 3 hours / week. Personal Study: 7 hours / week.
- 3 hours facilitated study consists of attending class, responding to facilitator feedback.
- Students are to allocate 7 hours of personal learning. This includes essential time spent on pre-reading and viewing materials, assessment progression and learning activities.

Learning Activities:

There are learning activities in each module with the types of activities listed in the delivery schedule above. These activities are an essential component of the subject and they are designed to prepare you for your graded assessment tasks. These learning activities will help you to build your understanding and skills whilst also collaborating with your peers.

***Detailed information of each activity can be found in their corresponding Module page in Blackboard.**